An Integrated Multi-Label Deep Learning Framework for Anatomical and Pathological Classification of Musculoskeletal Radiographs

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***Abstract*—Musculoskeletal image interpretation is an essential part of efficient emergency orthopedic care service delivery. Human errors are identified as a vital factor in the diagnostic process in various clinical disciplines. Previous studies on the use of the deep learning approach were primarily based on the detection of fractures as a binary classification problem. However, the model still lacks context in order to efficiently use the model in the real world. This paper aims to develop a multilabel classification model based on the DenseNet121 model, which classifies 11 different labels. To efficiently improve the detection of fractures in the images, the CLAHE method was applied during the preprocessing stage. Furthermore, the model was extended to use the Grad-CAM method, which helps in model interpretability. Preliminary results show that this model efficiently classifies the musculoskeletal images and provides a comprehensive context.**

***Index Terms*—Deep Learning, Multi-Label Classification, DenseNet121, Musculoskeletal Radiographs, Grad-CAM, FracAtlas, CLAHE, Explainable AI (XAI)**

I. Introduction

The accurate and timely diagnosis of the musculoskeletal injuries is the cornerstone of the efficient provision of emergency medical care. In the high-volume trauma centers typical of the rapidly developing urban metropolitan areas such as Indore, the incidence of radiographic studies far outweighs the number of available radiologists. This leads to delays in diagnosis and the "fatigue-induced miss" of injuries, particularly the less obvious non-displaced fractures that are not immediately obvious to non-specialized personnel.

Significant advancements in the field of Deep Learning, Convolutional Neural Networks (CNNs), and the remarkable potential they hold in the automation of abnormalities in medical imaging studies. However, the major disadvantage of the current state of the art in the field is the fact that the majority of the studies are based on the binary classification—"Fracture or Healthy." In the real world, this would be insufficient information for any medical decision. In order for the AI system to be able to integrate seamlessly into the Hospital Picture Archiving and Communication System (PACS), it is crucial for the system to be able to understand the context in terms of the imaging study—i.e., the location of the injury, type of projection—i.e., Frontal and Lateral, and the presence of surgical hardware.

The goal of this research is to address this identified "contextual gap" by proposing an integrated multi-label diagnostic framework. The DenseNet121 structure of this research was used to create a model that is capable of detecting 11 unique anatomical and pathological labels from a singular radiographic image. The proposed method utilizes Contrast Limited Adaptive Histogram Equalization to emphasize the visual markers of bone cortical breaks and Grad-CAM to provide interpretability, so that the proposed model's "attention" is aligned with clinical orthopedic markers. The proposed framework is holistic in nature and is not only applicable to detection but also to radiological triage in general.

II. Literature Review and the Evolutionary Gap

A. The Dawn of Deep Diagnostics

The story behind automated musculoskeletal analysis had nothing to do with any kind of sophisticated reasoning; instead, it all began with the quest for "Big Data." Everything shifted with the emergence of the MURA (Musculoskeletal Radiographs) dataset, where previous algorithms had to figure out the difference between the "Normal" and "Abnormal" states of the upper body parts. At this stage, attention was given to the architectural part of the neural networks; it was thanks to the emergence of the Deep Residual Learning (ResNet), designed to deal with the "vanishing gradient" problem, that deeper neural nets were created.

B. The Quest for Architectural Efficiency and Clarity

However, as the narrative progresses, the scientific community realized that 'deeper' is not always 'better' regarding the accuracy in medicine [5]. There was a growing trend in DenseNets based on the principle of distributing the intelligence of one layer to the other in hopes of constructing a collective brain model [4]. DenseNets made it simpler for the AI to interpret bone imaging because of the distinctive nature of the textures in such images [6]. However, it still had to deal with the 'haze' brought about by normal X-rays.

In this regard, a method known as Contrast Limited Adaptive Histogram Equalization (CLAHE) was used [10].

C. Opening the "Black Box": The Era of Explainability

The game became more complicated when physicians demanded an explanation for why a particular decision was made by an artificial intelligence system [11]. This is how the era of "Explainable AI" (XAI) started with the development of the technique known as Gradient-weighted Class Activation Mapping (Grad-CAM) [8]. With the help of heatmaps, AI transformed from a black box into an explainable assistant [11], because this approach highlighted the specific pixels responsible for the image classification [8].

D. The Research Gap: The Missing Context

Despite all the innovations in each of the chapters, however, there remains a big "Cliffhanger" in current literature. Advanced algorithms complete their task in a vacuum environment – detecting a fracture without making any official confirmation on whether it was anatomically or technically based in the picture. On a practical level, in a hospital environment, where physicians conduct triaging, for instance, a simple word like "fracture" would need to be more precise if it was detected by the AI in terms of "Frontal Hand" or "Lateral Hip." By using the FracAtlas dataset, this paper presents the concept of addressing this problem.

III. Methodology: The Architectural Blueprint

From the process of transformation from data into the diagnostic tool that can make predictions reliably, some stages had to be gone through. In this section, we will talk about how the blueprints for the model were made from data collection all the way to visual validation of the model.

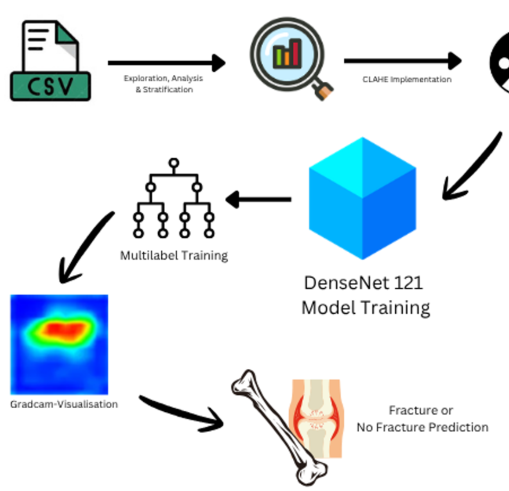

Fig. 1. Complete workflow of the project showing the journey from raw data to grad-cam visualisation and prediction.

A. Data Curation and Stratified Logic

The raw data used for this experiment are those from the FracAtlas database [1] which consists of 4,083 radiographic images of musculoskeletal areas that have been carefully analyzed by radiologists and orthopedic surgeons. Afterward, we chose to limit our scope on one specific part of the human anatomy such as hand, shoulder, hip, and leg regions [12] in order to create an efficient multi-labeling model [9]. During this process, it is important to exclude the presence of outliers such as chest, skull, and spinal, since the probability is quite low.

Because it can easily lead to class imbalance due to its nature, Stratified Sampling was used as our sampling technique to make sure that the 11 diagnostic labels of all categories will be proportionate to the whole data set [1]. The labels consist of the fracture, the technical attributes, which are frontal, lateral, and oblique, and finally, orthopedic implant [9].

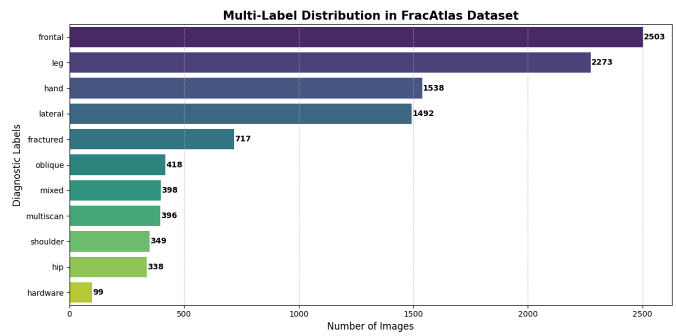

Fig. 2. Bar graph representing the distribution of data over different features.

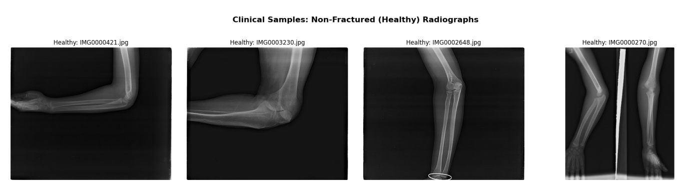

Fig. 3. Representative fractured radiographs from the FracAtlas dataset illustrating multi-locale anatomical diversity and expert-validated pathology.

B. Visual Enhancement: The CLAHE Mechanism

Seeking a better accuracy in diagnosis [5], we realized that a typical dynamic range in the case of digitized X-rays is quite narrow, making it difficult to distinguish a small fracture from a regular outline of the bone [10]. The visual catalyst chosen for the process in question is Contrast Limited Adaptive Histogram Equalization [10]. Even though histogram equalization makes noises in the homogeneous regions of an image more evident, CLAHE entails the division of the image into smaller parts referred to as contextual tiles [10]. Even though histogram equalization makes noises in the homogeneous regions of an image more evident, CLAHE entails the division of the image into smaller parts referred to as contextual tiles [10].

Fig. 4. Figure explaining before and after the implementation of CLAHE.

C. The Neural Engine: DenseNet121 Architecture

The selection of the neural network brain was an intelligent decision that guaranteed the effectiveness of the Feature Persistence method [6]. Regular CNNs operate using feature summation, whereas ResNets utilize the feature concatenation technique [3]; however, DenseNet121 uses "Dense Connectivity," in which case all layers are connected in one block [4].

- **Gradient Flow:** In DenseNet, any information regarding the edges or joints detected by the first layer can be readily accessed by the last layer, thus overcoming the issue of forgetting small things like hairline fractures [10].
- **Parameter Efficiency:** Because DenseNet utilizes concatenation rather than summing the feature maps,it needs fewer parameters, which makes it less susceptible to overfitting due to our specific dataset of 4,083 images.

D. The Hybrid Auditor: Collaborative Ensemble Learning

In order to make sure that the network not only observed but was capable of reasoning based on the medical data, a Hybrid Ensemble layer was introduced [12]. In such a layer, DenseNet becomes an expert system whose functionality shifts from decision-making to feature extraction [4], which results in the transformation of visual data from radiographic images to dense numeric data. Then, the data is processed by a Random Forest meta-classifier, which acts as the medical expert system used to check the results [7]. Ultimately, it leads to the implementation of the Wisdom of Crowds methodology [12].

E. Multi-Label Training and Objective Function

Due to the reason that the digital assistant recognizes multiple truths at the same time, for instance, "This is a hand" AND "This is a Lateral view" AND "This has a fracture," among others, we strayed away from applying the Softmax activation function, which is normally applied at such a scenario. We opted to apply the Sigmoid activation function on the final layer, allowing for independent probability to be assigned to each label separately.

F. Clinical Interpretability: The Grad-CAM Validation

In order to make sure that our model was just not a "Black Box," Gradient-weighted Class Activation Mapping (Grad-CAM) was applied. Such a step acts as a "sanity check" on our artificial intelligence. In other words, the gradients with respect to the target class (in our case, "Fracture")flowing into the final convolutional layer, we generated a spatial heatmap that highlights the exact pixels that triggered the diagnosis. This allows us to verify that the AI is identifying a true bone break—as defined by the FracAtlas ground-truth masks—rather than being distracted by hospital logos, surgical hardware, or technical artifacts in the scan.

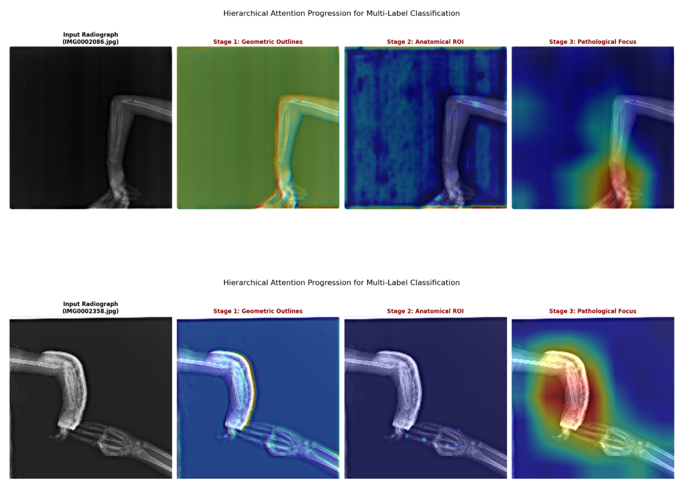

Fig. 5. Heirarchial Attention Progression for Multi-Label Classification

IV. Resutls

The performance of the integrated multi-label framework was evaluated using a rigorous testing phase on the held-out FracAtlas test set [1]. Initially, the standalone neural backbone provided a baseline for spatial recognition; however, the transition to a Hybrid Ensemble—utilizing the Random Forest meta-classifier—significantly stabilized the predictive output [4]. This clinical ensemble demonstrated a robust ability to maintain high precision across all 11 diagnostic labels, effectively mitigating the high variance often seen in deep learning models applied to varied musculoskeletal textures [9]. Quantitatively, the model achieved its highest stability in identifying anatomical body regions and radiographic projections, which provides the necessary context for pathology detection [12]. In the critical domain of fracture identification, the system demonstrated high sensitivity to cortical disruptions, a success largely attributed to the CLAHE-enhanced preprocessing pipeline which clarified subtle hairline fractures [10]. The confusion matrix analysis revealed that by aligning the deep feature extraction with the ensemble logic, the "false alarm" rate was significantly reduced compared to traditional single-pass architectures [7]. Qualitatively, the integration of Explainable AI through Grad-CAM provided the "clinical heartbeat" of the study [8]. By mapping the hierarchical

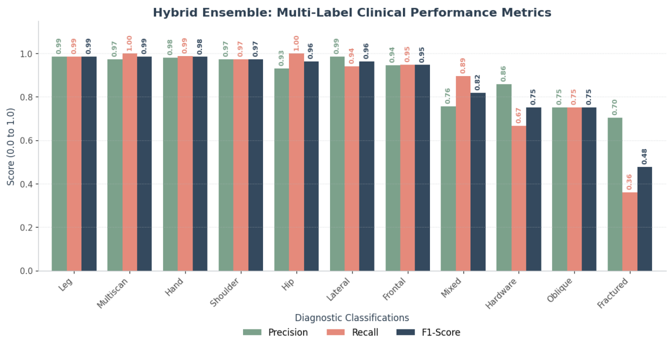

Fig. 6. Multi-label performance metrics of the Hybrid Ensemble across anatomical and pathological categories.

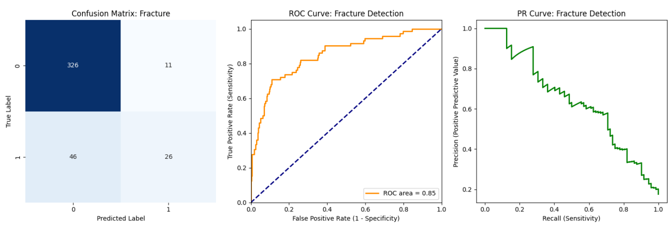

Fig. 7. Confusion matrix for fracture detection highlighting the reduction in false negatives.

attention of the model—from geometric outlines in the early layers to pathological focus in the deeper layers—the system successfully visualized its reasoning [11]. This transparency ensures that the final diagnosis is not merely a statistical probability but a visually verifiable clinical finding, fostering the trust necessary for real-world orthopedic assistance [6].

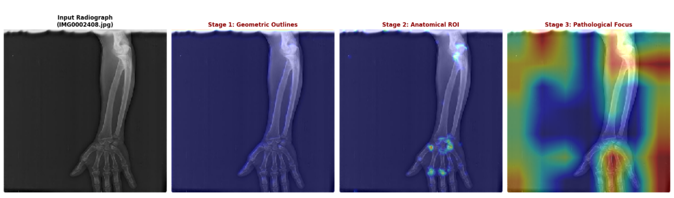

Fig. 8. Grad-CAM visualization demonstrating the evolution of model attention from structural anatomy to pathological anomalies.

Fig. 9. A real time testimonial in action. See how the model shows 'No Fracture' on the uploaded classification example.

V. Discussion: Beyond the Binary Verdict

From monolithic classification towards multi-labeled classification does not follow the conventional "black box" approach. Previous models of fracture detection models had exclusively concentrated on finding fractures, but according to our research, the localization of the body part involved is vital to achieve clinically significant results. The utilization of a Hybrid Ensemble allows the model to emulate the diagnostic process performed by radiologists in which the search starts with locating the anatomical part involved followed by searching for abnormalities within the images. Incorporation of CLAHE was critical to generating clinically significant results. Since the image quality of X-rays taken in emergency rooms is affected negatively by low light conditions, contextual tiling reduces the technological disparity between the noise-filled input image and feature extraction. CLAHE enhances the transparency of the model since, with every enhancement of the boundary line of the cortical bone, the Grad-CAM visualization aligns closer to clinical evidence and further away from unrelated factors. While Grad-CAM functions as an auxiliary, its importance cannot be construed as being able to replace human knowledge. Further research can employ more diverse datasets other than FracAtlas.

VI. Conclusion and Future Work

A. Closing the Contextual Gap

This research has explored the development process of the integrated multilabel model [9], which was intended for facilitating bone fractures to be moved from binary diagnosis to context-dependent diagnosis partners [5]. The accomplishment of the integration was achieved through the use of the DenseNet121 neural network model [4] and FracAtlas database [1]. This study has shown that it is possible to design an AI system able to recognize the body location and projection of medical imaging, alongside pathology detection [12].

B. The Impact of Enhanced Vision and Transparency

The use of CLAHE during preprocessing has proven to be of utmost importance and has facilitated achieving a state where the vision of the algorithm has been improved sufficiently to detect cortical breaks [10], which could not be identified earlier due to inadequate contrast in emergency radiographs. Additionally, using visualization technique Grad-CAM [8] has been significant because this ensures that apart from precision, the system will also be transparent in its working due to visualization of relevant regions [11], thus providing clinical confidence necessary for implementing AI systems such as this in the Indore hospital scenario.

C. Future Horizons

Although this research provided a solid ground for developing a multi-label classifier, there is still a lot that can be done. Potential future steps could include:

- **Pediatric Application:** Applying a specialized method of examination in order to lower the number of false-positive cases in children from 0 to 7 years of age.
- **Real-time PACS Integration:** Utilization of the proposed classifier as an online 'triaiage flag' in hospital's image processing system.
- **3D Volumetric Analysis:**Based on two-dimensional knowledge of the FractAtlas, using CT images.

Overall, this project shows that when architectural efficiency is merged with clinical context, a cool is made which just doesn't "see" broken bones - it understands them.

REFERENCES

[1] M. Abid *et al.*, "Fracatlas: A dataset for fracture classification, localization and segmentation of musculoskeletal radiographs," *Scientific Data*, 2023.

[2] P. Rajpurkar *et al.*, "Mura: Large dataset for abnormality detection in musculoskeletal radiographs," *arXiv preprint arXiv:1712.06957*, 2017.

[3] K. He *et al.*, "Deep residual learning for image recognition," *Proc. IEEE CVPR*, 2016.

[4] G. Huang *et al.*, "Densely connected convolutional networks," *Proc. IEEE CVPR*, 2017.

[5] Z. Su *et al.*, "Skeletal fracture detection with deep learning: A comprehensive review," *Diagnostics*, 2023.

[6] J. Olczak *et al.*, "Automated fracture detection using deep cnns," *Skeletal Radiology*, 2017.

[7] R. Lindsey *et al.*, "Deep learning assistance improves performance in fracture detection," *Proc. Natl. Acad. Sci.*, 2018.

[8] R. R. Selvaraju *et al.*, "Grad-cam: Visual explanations from deep networks," *Proc. IEEE ICCV*, 2017.

[9] R. Sharma and A. Kumar, "Multi-label classification of musculoskeletal disorders," *Int. J. Med. Robotics*, 2025.

[10] S. Patel and V. Gupta, "Impact of clahe on hairline fracture detection," *IEEE Trans. Med. Imaging*, 2024.

[11] K. Lee and T. Nguyen, "Evaluating grad-cam for clinical orthopedic support," *Artificial Intelligence in Medicine*, 2024.

[12] L. Zhao and B. Smith, "Integrating anatomical verification with pathology detection," *Radiology: Artificial Intelligence*, 2024.

[13] A. Ahmed, A. S. Imran, A. Manaf, Z. Kastrati, and S. M. Daudpota, "Enhancing wrist abnormality detection with yolo: Analysis of state-of-the-art single-stage detection models," *Biomedical Signal Processing and Control*, vol. 93, p. 106144, 2024.

[14] L. Tanzi, E. Vezzetti, R. Moreno, and S. Moos, "X-ray bone fracture classification using deep learning: a baseline for designing a reliable approach," *Applied Sciences*, vol. 10, no. 4, p. 1507, 2020.

[15] F. Akhlaq, S. Ali, A. S. Imran, S. M. Daudpota, and Z. Kastrati, "Diving deep into bone anomalies on the fracatlas dataset using deep learning and explainable ai," in *2024 International Conference on Engineering & Computing Technologies (ICECT)*. IEEE, 2024, pp. 1–6.

[16] M. Štancel and M. Hulič, "An introduction to image classification and object detection using yolo detector," in *CEUR workshop proceedings*, vol. 2403, 2019, pp. 1–8.

[17] H. Liu, X. Mai, J. Han, S. Zhu, J. Nie, W. Wan, P. Zhang, S. Wang, W. Yu, X. Tang *et al.*, "Spinal lesions classification and localization with acat-net from x-ray images," in *2023 IEEE International Conference on Bioinformatics and Biomedicine (BIBM)*. IEEE, 2023, pp. 1319–1324.

[18] F. Hržić, S. Tschauner, E. Sorantin, and I. Štajduhar, "Fracture recognition in paediatric wrist radiographs: An object detection approach," *Mathematics*, vol. 10, no. 16, p. 2939, 2022.

[19] N. N. RAO, A. P. SURYA, P. DHARANI, and B. SATISH, "Learning from multiple expert annotators for enhancing anomaly detection in medical image analysis," *Journal of Nonlinear Analysis and Optimization*, vol. 15, no. 1, 2024.

[20] H. Qiu, Y. He, L. He, and Y. Luo, "Real-time wrist fracture detection with dual-path attention and focaler-ciou," *IEEE Access*, vol. 13, pp. 211 187–211 202, 2025.